

# Fused Routed Decode Attention (compute decode\_attention\_static, mx.compile)

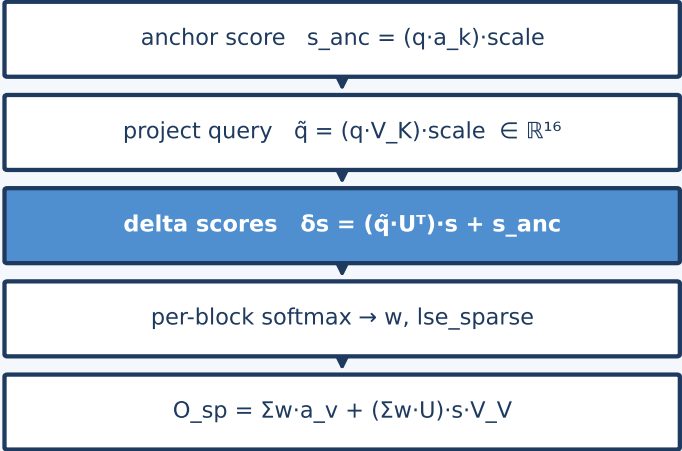
Route top-K blocks by EXACT residual-key score  $\rightarrow$  low-rank branch (compute decode\_attention\_static, mx.compile) + exact residuals + recency  $\rightarrow$  LSE merge

query  $q$  [H, d]

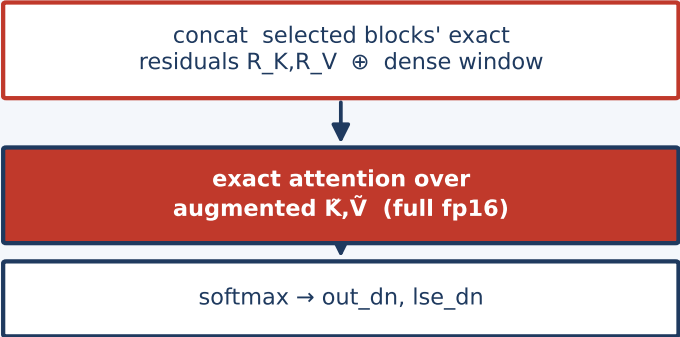
operator:  $p_b = \max(q \cdot a_k, \max_j q \cdot R_{K,j}) \cdot \text{scale} \rightarrow \text{keep top-K} = 16 \text{ blocks}$

exact q-k over  
anchor + residual keys

## Low-rank branch (selected blocks)



## Exact branch (residuals + recency)



verbatim recall of needle tokens

$O(K \cdot r \cdot B)$  — no  $O(B \cdot d)$  decompression

flash-style LSE merge  
 $m = \max(lse\_sp, lse\_dn)$   
 $out = (e^{\dots} O_{sp} + e^{\dots} O_{dn}) / \Sigma$

NaN/inf-guarded so an empty compressed set contributes exactly zero weight